

Project Proposal: Robust & Interpretable Brain Tumor Classification Using Sharpness-Aware Minimization

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CS 518: Deep Learning for Computer Vision

Spring 2026

1. Dataset Description & Clinical Motivation

The project will utilize the Brain Tumor MRI Dataset aggregated on Kaggle, which combines data from several open-source medical imaging benchmarks (including Figshare, SARTAJ, and Br35H). The dataset contains approximately 7,023 human brain MRI images organized into four distinct classes: Glioma, Meningioma, Pituitary Tumor, and No Tumor.

Automating this classification task is clinically vital, as different tumor types require drastically different intervention strategies. A robust model can serve as a diagnostic "second opinion," reducing radiologist fatigue. Furthermore, the 2D image formats (JPG/PNG) bypass the computational overhead of 3D voxel processing, allowing us to focus entirely on advanced algorithmic optimization and interpretability.

2. Task and Problem Formulation

(a) Pre-processing & Augmentation: Raw MRI scans inherently vary in resolution and hardware-induced noise. The pre-processing pipeline will first resize all images to a fixed spatial dimension of 224×224 pixels. Min-Max Normalization will be applied to scale pixel intensities to the $[0, 1]$ range, and a Gaussian blur will be used to smooth artifacts.

Medical datasets are prone to overfitting because the background remains visually static across samples. To combat this, we will implement domain-specific augmentation: random horizontal flips, slight rotations ($\pm 10^\circ$), and brightness adjustments. We will strictly avoid "shearing," which could unrealistically warp tumor shapes and destroy clinically relevant morphological features.

(b) Main Task: The objective is Multi-Class Image Classification where the input $x_i \in \mathbb{R}^{H \times W \times C}$ is a 2D MRI slice, and the output $y_i \in \{1, 2, 3, 4\}$ is the predicted tumor category. The dataset will be split into Training (70%), Validation (15%), and Testing (15%) sets to prevent data leakage.

3. Deep Learning Pipeline & Loss Formulation

We propose using the **EfficientNet-B0** architecture as our primary feature extractor, alongside a **ResNet-50** baseline for comparative analysis. The grayscale MRIs will be replicated across 3 channels to fully leverage ImageNet weights. Features will pass through a Global Average Pooling layer, a Dropout layer ($p = 0.5$), and into a 4-neuron dense classification head.

To handle class imbalance, we will use **Focal Loss**, which dynamically scales the loss based on prediction confidence by adding a modulating factor $(1 - p_t)^\gamma$ to the cross-entropy criterion. This forces the network to focus its learning capacity on the hardest-to-predict, minority tumor types.

4. Advanced Optimization: Sharpness-Aware Minimization (SAM)

The core research component is **Sharpness-Aware Minimization (SAM)**. Standard optimizers often converge to "sharp" minima that generalize poorly to unseen medical data. SAM improves generalization by actively seeking parameters θ whose entire neighborhood has uniformly low loss, formulated as a minimax problem:

$$\min_{\theta} \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}(\theta + \epsilon) \quad (1)$$

Where ρ is the neighborhood size hyperparameter, and ϵ is the adversarial perturbation.

Implementing SAM requires a custom, two-step PyTorch training loop:

1. **The Ascent Step:** Standard gradients calculate the perturbation ϵ , moving the weights to a nearby point where the loss is maximized (the "worst-case" scenario).
2. **The Descent Step:** A second pass is executed at this perturbed position. The optimizer updates the original weights using these new gradients, forcing the model into a flat valley of low loss.

5. Post-Hoc Explainability: Grad-CAM & Integrated Gradients

In medical AI, clinicians must trust that the model makes decisions based on pathology, not image artifacts. We will integrate **Gradient-weighted Class Activation Mapping (Grad-CAM)**.

Grad-CAM utilizes the gradients of the predicted tumor class flowing into the final convolutional layer. The neuron importance weights α_k^c for a specific class c and feature map k are calculated via global average pooling:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial Y^c}{\partial A_{ij}^k} \quad (2)$$

We perform a weighted combination of forward activation maps, followed by a ReLU activation, to generate a spatial heatmap that is overlaid onto the MRI scan. We will cross-reference these heatmaps using **Integrated Gradients** to ensure explanation stability. This visually proves the network's features are localized precisely on the tumor boundaries.

6. Project Scope & Division of Labor

This project is designed for a team of four, with responsibilities strictly divided for parallel progress:

- **Member 1 (Data & Preprocessing):** Custom PyTorch DataLoader, normalization, blurring, and the medically-safe data augmentation pipeline.
- **Member 2 (Architecture & Baseline):** Instantiation of EfficientNet-B0 and ResNet-50, Focal Loss integration, and standard baseline training loop construction.
- **Member 3 (Optimization):** Custom two-step PyTorch training loop for SAM, including hyperparameter tuning for the neighborhood radius ρ .
- **Member 4 (Explainability & Evaluation):** Grad-CAM and Integrated Gradients visualization pipelines, and evaluation metrics (Macro-F1, Precision, Recall, AUROC, Confusion Matrices).

References

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